Work Breakdown Structure(WBS) and Deliverables

Deliverables:

Module 1:

Product Deliverables:

1. User Registration and Authentication System
2. User Profile Management System
3. Portfolio Management System
4. Work History Tracking Feature
5. Review and Rating System
6. Database Design and Implementation
7. APIs for Integration
8. User Interface Components
9. Testing and Validation Artifacts
    - 9.1. Test Cases
    - 9.2. Test Results
    - 9.3. Bug Reports

Process Deliverables:

1. Sub-Project Management Plan (SPMP)
2. Work Breakdown Structure (WBS)
3. Schedule Baseline and Gantt Chart
4. Milestone List
5. Progress Reports
6. Requirements Management Plan (SP-RMP)
7. Software Requirements Specification (SP-SRS)
8. Software Design Specification (SP-SDS)
9. Testing Plan
10. Final Project Report and Demonstration

Module 2:

Product Deliverables:

1. Online testing system
2. Question bank database
3. Automated grading engine
4. Certification system
5. Test history dashboard

Process Deliverables:

1. SPMP
2. SP-SRS
3. SP-SDS

Module 3:

Product Deliverables:

1. Job Posting System
2. Freelancer Bidding and Proposal System
3. Gig/Service Listing System
4. Advanced Search and Filtering Engine
5. Job and Bid Status Management Workflows
6. Client Dashboard
7. Freelancer Dashboard
8. Admin Moderation Interface
9. RESTful APIs for Module Integration
10. Relational Database Schema

Process Deliverables:

1. SPMP (Sub-Project Management Plan)
2. SP-SRS (Software Requirements Specification)
3. SP-RMP (Requirements Management Plan)
4. SP-SDS (Software Design Specification)
5. API Documentation
6. Test Cases and Testing Reports
7. User Manual

Module 4:

Product Deliverables:

1. Matching Engine
2. Match Percentage & Ranking System
3. Freelancer Interface (Recommended Jobs, Match Scores, Trending Skills)
4. Client Interface (Top Matched Freelancers, Ranked Candidates)
5. Notification System (Skill Gap Alerts & Suggestions)

Process Deliverables:

1. Requirements Specification
2. System & Database Design
3. API & UI/UX Design
4. Implementation & Integration Plan
5. Testing (Test Cases & Reports)
6. Technical Documentation
7. Final Report & Demonstration

Module 5:

Product Deliverables:

1. V1.0 (MVP): Functional RBAC and team formation features

2. V2.0 (Full Feature): Task management system, workspace dashboard, and activity logging

Process Deliverables:

1. SP-SRS: Software Requirements Specification
2. SP-SDS: Software Design Specification
3. SP-RMP: Risk Management Plan
4. Sprint Burn-down Charts
5. Definition of Done (DoD) Checklist

Module 6:

Product Deliverables:

1. Messaging System
2. Group Chat System
3. File Sharing System4. Notification System
4. Notification System
5. Meeting Integration Feature
6. Message History Database
7. User Interface Design
8. Test Plan

Process Deliverables:

1. Sub-Project Management Plan (SPMP)
2. Requirements Management Plan (SP-RMP)
3. Software Requirements Specification (SP-SRS)
4. Software Design Specification (SP-SDS)
5. Project charter
6. Scope statement
7. Work Breakdown Structure (WBS)
8. Schedule baseline
9. Milestones list
10. Progress and status reports
11. Risk management documentation
12. Final project report

Module 7:

Product Deliverables:

1. Escrow Payment System for secure holding of client funds
2. Milestone-Based Payment Processing system
3. Invoice Generation system for completed work

4. Payment Release mechanism after milestone approval
5. Freelancer Withdrawal system to supported wallets or bank accounts
6. Transaction History tracking for all users
7. Multiple Currency Support for international transactions
8. Integration with at least three wallets or banking services

Process Deliverables:

1. Sub-Project Management Plan (SPMP) for the Payment module
2. Requirements Management Plan (SP-RMP)
3. Software Requirements Specification (SP-SRS)
4. Software Design Specification (SP-SDS)
5. Testing and validation reports for payment transactions
6. Integration documentation with other platform modules

Module 8:

Product Deliverables:

1. Dispute Submission System
2. Evidence Management System
3. Dispute Lifecycle Tracking
4. Mediation Interface
5. Admin Arbitration System
6. Resolution Reporting System
7. Admin Profile Management System
8. Database Design and Implementation
9. APIs for Integration
10. User Interface Components
11. Testing and Validation Artifacts

Process Deliverables:

1. Sub-Project Management Plan (SPMP)
2. Work Breakdown Structure (WBS)
3. Schedule Baseline and Gantt Chart
4. Milestone List
5. Progress Reports
6. Requirements Management Plan (SP-RMP)
7. Software Requirements Specification (SP-SRS)
8. Software Design Specification (SP-SDS)
9. Testing Plan
10. Final Project Report and Demonstration

Module 11:

Product Deliverables:

1. Points Economy & Progression System
    a. API for awarding points
    b. Multi-tier leveling system (Levels 1–3)
    c. Points ledger
2. Achievement Engine
    a. Rules engine for badge awarding
    b. Automatic badge assignment
3. Gamified Onboarding
    a. 5-step onboarding checklist
4. Challenges System
    a. Daily and weekly challenges
    b. Challenge tracking interface
5. Leaderboard
    a. Ranking system with tiebreaker logic
6. Trust Score Module
    a. Trust score calculation logic
7. Notification System
    a. In-app notification panel
8. Final Integrated Module
    a. Complete tested integrated codebase

Process Deliverables:

1. Project Management Plan (PMP)
2. Software Requirements Specification (SRS) v1.0
3. Software Design Specification (SDS)
4. Risk Management Plan (RMP)
5. Project Scope Statement
6. Work Breakdown Structure (WBS) & WBS Dictionary
7. Requirements Traceability Matrix (RTM)
8. Postman Collection

WBS:

1. National Freelance And Skill Verification System
    - 1.1. User Identity, Profile & Portfolio Management Module
        - 1.1.1. Project Management
            - 1.1.1.1. Planning
            - 1.1.1.2. Meetings & Coordination
            - 1.1.1.3. Progress Tracking & Reporting
        - 1.1.2. Requirements Analysis
            - 1.1.2.1. Requirement Gathering
            - 1.1.2.2. Requirement Analysis
            - 1.1.2.3. Requirement Documentation (SRS)
        - 1.1.3. System Design
            - 1.1.3.1. System Architecture Design
            - 1.1.3.2. Database Design (User, Profile, Portfolio)
            - 1.1.3.3. API Design
            - 1.1.3.4. UI/UX Design
        - 1.1.4. Development – User Identity Management
            - 1.1.4.1. User Registration Implementation
            - 1.1.4.2. Authentication (Login/Logout)
            - 1.1.4.3. Role-Based Access Control
        - 1.1.5. Development – Profile Management
            - 1.1.5.1. Profile Creation Feature
            - 1.1.5.2. Profile Editing Feature
            - 1.1.5.3. Profile Viewing Feature
        - 1.1.6. Development – Portfolio Management
            - 1.1.6.1. Portfolio Upload Feature
            - 1.1.6.2. Portfolio Edit/Delete Feature
            - 1.1.6.3. Portfolio Display Interface
        - 1.1.7. Development – Work History & Reviews
            - 1.1.7.1. Work History Tracking
            - 1.1.7.2. Review & Rating System
        - 1.1.8. Integration & API Development
            - 1.1.8.1. API Implementation
            - 1.1.8.2. Integration with Other Modules
        - 1.1.9. Testing
            - 1.1.9.1. Unit Testing
            - 1.1.9.2. Integration Testing
            - 1.1.9.3. System Testing
            - 1.1.9.4. Bug Fixing
        - 1.1.10. Documentation
            - 1.1.10.1. SP-SRS Preparation
            - 1.1.10.2. SP-SDS Preparation
            - 1.1.10.3. Technical Documentation

- 1.1.11. Deployment & Demonstration
  - 1.1.11.1. Deployment Setup
  - 1.1.11.2. Final Demonstration
  - 1.1.11.3. Final Report Submission
- 1.2. Skill Testing and Certification
  - 1.2.1. Design
    - 1.2.1.1. System Architecture
    - 1.2.1.2. Database Design
    - 1.2.1.3. UI Design
  - 1.2.2. Development
    - 1.2.2.1. Question Bank
    - 1.2.2.2. Exam Engine
    - 1.2.2.3. Grading System
    - 1.2.2.4. Certification Module
    - 1.2.2.5. History Tracking
  - 1.2.3. Integration
    - 1.2.3.1. API Integration
    - 1.2.3.2. Module Connectivity
  - 1.2.4. Testing
    - 1.2.4.1. Unit Testing
    - 1.2.4.2. Integration Testing
- 1.3. Project & Gig Marketplace Module
  - 1.3.1. Project Management
    - 1.3.1.1. Planning & Scheduling
    - 1.3.1.2. Progress Monitoring
    - 1.3.1.3. Team Coordination & Meetings
  - 1.3.2. Product Requirements
      - 1.3.2.1.1. Functional Requirements
      - 1.3.2.1.2. Job Posting Requirements
      - 1.3.2.1.3. Bidding System Requirements
      - 1.3.2.1.4. Gig/Service Creation Requirements
      - 1.3.2.1.5. Search & Filter Requirements
      - 1.3.2.1.6. Job Status Management Requirements
      - 1.3.2.1.7. Project Categories Requirements
    - 1.3.2.2. Non-Functional Requirements
      - 1.3.2.2.1. Performance & Scalability
      - 1.3.2.2.2. Security Requirements
      - 1.3.2.2.3. Usability Requirements
      - 1.3.2.2.4. Availability & Reliability
    - 1.3.2.3. Requirements Review & Sign-off
  - 1.3.3. Detail Design
    - 1.3.3.1. Software Design
      - 1.3.3.1.1. Job Database Schema
      - 1.3.3.1.2. Bidding System Architecture

1.3.3.1.3. Search Engine Design
1.3.3.1.4. API Interface Definitions
1.3.3.2. User Documentation (Design Phase)
1.3.3.2.1. UI/UX Wireframes
1.3.3.2.2. Data Flow Diagrams
1.3.3.3. Design Review & Approval
1.3.4. Construct
1.3.4.1. Core Feature Development
1.3.4.1.1. Job Posting Feature
1.3.4.1.2. Freelancer Bidding System
1.3.4.1.3. Gig/Service Creation Feature
1.3.4.1.4. Search Engine & Filters
1.3.4.1.5. Job Status Management
1.3.4.1.6. Project Categories Feature
1.3.4.2. User Documentation (Development Phase)
1.3.4.2.1. Developer Notes
1.3.4.2.2. API Documentation
1.3.4.3. Code Review & Unit Testing
1.3.5. Integration and Test
1.3.5.1. Integration & System Testing
1.3.5.1.1. Integration with Auth/User Module
1.3.5.1.2. Integration with Payment Module
1.3.5.1.3. Integration with AI Matching Module
1.3.5.1.4. System Testing
1.3.5.2. User Documentation (Final)
1.3.5.2.1. User Manual
1.3.5.2.2. Final API Docs
1.3.5.3. Bug Fixes & Regression Testing
1.3.5.4. Final Handoff & Closure
1.4. AI Matching & Recommendation Module
1.4.1. Planning Deliverables
1.4.1.1. Requirements Documentation
1.4.1.2. Stakeholder Analysis Report
1.4.1.3. Project Documentation
1.4.2. Design Deliverables
1.4.2.1. System Architecture
1.4.2.2. Database Design (Skills, Projects, Matching Data)
1.4.2.3. API Integration Design
1.4.3. Development Deliverables
1.4.3.1. Skill Matching Engine
1.4.3.2. Budget & Rate Filtering System
1.4.3.3. Match Percentage Calculation System
1.4.3.4. Skill Gap Analysis System
1.4.3.5. Trending Skills Aggregation System

- 1.4.3.6. Notification System Integration
- 1.4.4. Testing Deliverables
  - 1.4.4.1. Unit Test Results
  - 1.4.4.2. API Test Results
  - 1.4.4.3. Performance Test Results
- 1.4.5. Deployment Deliverables
  - 1.4.5.1. Integrated Module (API Level)
  - 1.4.5.2. Final Validated & Deployed System
- 1.4.6. Documentation Deliverables
  - 1.4.6.1. Requirements Management Plan (RMP)
  - 1.4.6.2. Software Requirements Specification (SRS)
  - 1.4.6.3. Software Design Specification (SDS)

1.5. Collaboration And Team Workspace

- 1.5.1. Project Management & Integration Planning Deliverables
  - 1.5.1.1. Agile Sprint Plan & Product Backlog
  - 1.5.1.2. Inter-Module Interface Specification
  - 1.5.1.3. API Boundaries Definition (Inter-Module Coordination)
  - 1.5.1.4. Project Management Plan (PMP) & Risk Assessment
- 1.5.2. Requirements & Architectural Design Deliverables
  - 1.5.2.1. Software Requirements Specification (SP-SRS)
  - 1.5.2.2. Software Design Specification (SP-SDS)
  - 1.5.2.3. Database Schema for Workspaces & Task Boards
  - 1.5.2.4. System Architecture for Workspace Management & Task Processing
  - 1.5.2.5. RBAC Permission Matrix (Admin, Client, PM, Freelancer)
- 1.5.3. Foundation & Workspace Security Deliverables
  - 1.5.3.1. Team Formation System (Invitation & Accept/Decline Workflows)
  - 1.5.3.2. Server-Side RBAC System (Role-Validation Middleware)
  - 1.5.3.3. Workspace Data Isolation Mechanism
- 1.5.4. Task Management Deliverables
  - 1.5.4.1. Task Creation & Assignment System
  - 1.5.4.2. Task Status Management System (To Do, In Progress, Under Review, Completed)
  - 1.5.4.3. Task Editing & Deletion System
  - 1.5.4.4. Task Board Interface
- 1.5.5. Monitoring & Dashboard Deliverables
  - 1.5.5.1. Activity Logging System (Member Actions, Task Updates, Role Changes)
  - 1.5.5.2. Workspace Dashboard (Project Details, Team Members, Progress Tracking)
  - 1.5.5.3. Automatic Dashboard Update Mechanism
- 1.5.6. Testing, Quality Assurance & Integration Deliverables
  - 1.5.6.1. Uptime & Reliability Test Results (99.5% Availability)
  - 1.5.6.2. Performance Audit Report (≤ 3 Seconds Response Time)

1.5.6.3. User Acceptance Testing (UAT) Results (Cross-Browser Validation)
1.5.6.4. Integrated System with National Platform Core

1.6. Communication
1.6.1. Project Management Deliverables
1.6.1.1. Project Plan
1.6.1.2. Meeting Records & Coordination Logs
1.6.1.3. Progress Reports
1.6.2. Requirements Deliverables
1.6.2.1. Requirements Management Plan (SP-RMP)
1.6.2.2. Requirements Specification (SP-SRS)
1.6.2.3. Requirements Analysis Report
1.6.3. System Design Deliverables
1.6.3.1. System Architecture
1.6.3.2. Database Schema (Messages, Files, Chat History)
1.6.3.3. API Design Specification
1.6.3.4. UI/UX Design Mockups
1.6.4. Messaging System Deliverables
1.6.4.1. Real-time Messaging System
1.6.4.2. Message Delivery & Synchronization
1.6.5. Group Chat System Deliverables
1.6.5.1. Group Creation & Management
1.6.5.2. Multi-user Messaging
1.6.6. File Sharing System Deliverables
1.6.6.1. File Upload & Sharing
1.6.6.2. Secure File Storage
1.6.7. Notification System Deliverables
1.6.7.1. Real-time Notification System
1.6.7.2. Notification Trigger Mechanism
1.6.8. Meeting Integration Deliverables
1.6.8.1. Meeting Link Generation
1.6.8.2. Meeting Link Sharing
1.6.9. Message History & Database Deliverables
1.6.9.1. Message Storage System
1.6.9.2. Chat History Retrieval
1.6.10. User Interface Deliverables
1.6.10.1. Chat Interface Screens
1.6.10.2. Notification Panels
1.6.10.3. User Interaction Flows
1.6.11. Integration Deliverables
1.6.11.1. Messaging API Services
1.6.11.2. Integration with User Module
1.6.11.3. Integration with Project/Collaboration Modules
1.6.12. Testing Deliverables

1.6.12.1. Unit Test Results
1.6.12.2. Integration Test Results
1.6.12.3. System Test Reports
1.6.12.4. Bug Reports & Fix Logs
1.6.13. Documentation Deliverables
1.6.13.1. SP-SRS Document
1.6.13.2. SP-SDS Document
1.6.13.3. Technical Documentation
1.6.14. Deployment & Final Deliverables
1.6.14.1. Deployed Communication Module
1.6.14.2. Final Demonstration
1.6.14.3. Final Report
1.7. Payment And Escrow Management
1.7.1. Project Management Deliverables
1.7.1.1. Payment Module Scope Definition
1.7.1.2. Payment Features Identification
1.7.1.3. Module Boundaries Definition
1.7.1.4. Stakeholder Identification (Internal & External)
1.7.1.5. Software Project Management Plan (SPMP)
1.7.1.5.1. Executive Summary
1.7.1.5.2. Scope Definition
1.7.1.5.3. Work Breakdown Structure (WBS)
1.7.2. Requirements Deliverables
1.7.2.1. Functional Requirements (Escrow, Milestone Payments, Invoices, Withdrawals)
1.7.2.2. Non-Functional Requirements (Security, Performance, Reliability)
1.7.2.3. Payment Workflow Definitions
1.7.2.3.1. Client Payment Flow
1.7.2.3.2. Freelancer Withdrawal Flow
1.7.2.3.3. Refund Flow
1.7.2.4. Requirements Documentation
1.7.2.4.1. SP-RMP
1.7.2.4.2. SP-SRS
1.7.3. System Design Deliverables
1.7.3.1. Escrow System Design
1.7.3.1.1. Fund Holding Mechanism
1.7.3.1.2. Payment Release Logic
1.7.3.2. Milestone Payment Design
1.7.3.2.1. Milestone Tracking
1.7.3.2.2. Payment Triggers
1.7.3.3. Database Design
1.7.3.3.1. Transaction Tables
1.7.3.3.2. User Wallet Records
1.7.3.4. Wallet/Bank Integration Design

1.7.3.4.1. API Interface Design
1.7.3.4.2. Currency Handling Design
1.7.3.5. Software Design Specification (SP-SDS)
1.7.4. Development Deliverables
1.7.4.1. Escrow System
1.7.4.1.1. Client Fund Storage
1.7.4.1.2. Fund Locking Mechanism
1.7.4.2. Milestone Payment System
1.7.4.2.1. Milestone Tracking Logic
1.7.4.2.2. Payment Release Triggers
1.7.4.3. Invoice System
1.7.4.3.1. Invoice Generation
1.7.4.3.2. Invoice Records Storage
1.7.4.4. Payment Release System
1.7.4.4.1. Payment Approval
1.7.4.4.2. Fund Transfer
1.7.4.5. Withdrawal System
1.7.4.5.1. Withdrawal Request Handling
1.7.4.5.2. Withdrawal Processing
1.7.4.6. Transaction History System
1.7.4.6.1. Transaction Records
1.7.4.6.2. Transaction History Display
1.7.4.7. Multi-Currency Support
1.7.4.7.1. Currency Conversion Logic
1.7.4.7.2. Currency Storage
1.7.5. Integration Deliverables
1.7.5.1. User Profile Module Integration
1.7.5.1.1. User Wallet Linking
1.7.5.2. Marketplace Module Integration
1.7.5.2.1. Payment Initiation from Projects
1.7.5.3. Dispute Resolution Module Integration
1.7.5.3.1. Refund Handling
1.7.5.4. Analytics Module Integration
1.7.5.4.1. Transaction Data Reporting
1.7.6. Testing & Validation Deliverables
1.7.6.1. Transaction Test Results
1.7.6.1.1. Payment Success Cases
1.7.6.1.2. Payment Failure Cases
1.7.6.2. Security Validation Results
1.7.6.2.1. Fraud Detection Checks
1.7.6.2.2. Data Protection Validation
1.7.6.3. Integration Test Results
1.7.6.3.1. Module Interaction Testing
1.7.6.4. Bug Fixes & Final Validation Results

1.8. Dispute & Conflict Resolution + Admin Management Module – Deliverable-Based WBS

- 1.8.1. Project Management Deliverables
  - 1.8.1.1. Project Plan
  - 1.8.1.2. Meeting Records & Coordination Logs
  - 1.8.1.3. Progress Reports
- 1.8.2. Requirements Deliverables
  - 1.8.2.1. Requirements Specification (SRS)
  - 1.8.2.2. Requirements Analysis Report
  - 1.8.2.3. Requirements Documentation
- 1.8.3. System Design Deliverables
  - 1.8.3.1. System Architecture
  - 1.8.3.2. Database Schema (Disputes, Evidence, Admin)
  - 1.8.3.3. API Design Specification
  - 1.8.3.4. UI/UX Design Mockups
- 1.8.4. Dispute Management System Deliverables
  - 1.8.4.1. Dispute Submission System
  - 1.8.4.2. Evidence Storage System
  - 1.8.4.3. Dispute Status Tracking System
- 1.8.5. Mediation & Arbitration Deliverables
  - 1.8.5.1. Mediation Interface
  - 1.8.5.2. Admin Arbitration Panel
  - 1.8.5.3. Resolution Decision Records
- 1.8.6. Resolution & Reporting Deliverables
  - 1.8.6.1. Resolution Reports
  - 1.8.6.2. Notification System
  - 1.8.6.3. Resolution History Records
- 1.8.7. Admin Management Deliverables
  - 1.8.7.1. Admin Authentication System
  - 1.8.7.2. Role & Permission Management System
  - 1.8.7.3. Admin Dashboard
- 1.8.8. Integration Deliverables
  - 1.8.8.1. API Services
  - 1.8.8.2. User Module Integration
  - 1.8.8.3. Project/Payment/Communication/Notification Integrations
- 1.8.9. Testing Deliverables
  - 1.8.9.1. Unit Test Results
  - 1.8.9.2. Integration Test Results
  - 1.8.9.3. System Test Reports
  - 1.8.9.4. Bug Reports & Fix Logs
- 1.8.10. Documentation Deliverables
  - 1.8.10.1. SP-SRS Document
  - 1.8.10.2. SP-SDS Document
  - 1.8.10.3. Technical Documentation

1.8.11. Deployment & Final Deliverables
1.8.11.1. Deployed System
1.8.11.2. Final Demonstration
1.8.11.3. Final Report
1.9. Analytics And System Governance
1.9.1. Sub-Project Management Deliverables
1.9.1.1. SPMP Scope Definition & WBS
1.9.1.2. Project Schedule & Gantt Chart
1.9.1.3. Scope & Schedule Management Plans
1.9.2. Requirements Deliverables
1.9.2.1. Dashboard UI Requirements
1.9.2.2. Identified Data Sources from Core Modules
1.9.2.3. Analytics Categories Definition
1.9.3. System Design Deliverables
1.9.3.1. Dashboard UI/UX Design
1.9.3.2. Data Aggregation Architecture
1.9.3.3. Read-Only Database Query Designs
1.9.3.4. Data Privacy Rules
1.9.3.5. SP-SDS Document
1.9.4. Software Development Deliverables
1.9.4.1. Admin Dashboard UI
1.9.4.2. Data Aggregation Engine
1.9.4.3. KPI Calculation Logic
1.9.5. Testing & Quality Assurance Deliverables
1.9.5.1. Backend Data Accuracy Test Results
1.9.5.2. Frontend UI/UX Validation Results
1.9.6. Integration & Deployment Deliverables
1.9.6.1. Integrated Dashboard in Admin Portal
1.9.6.2. System Integration Test Results
1.9.6.3. Final Sub-Project Review & Sign-Off
1.10. Social Impact And Volunteering Exchange
1.10.1. No Submission from the Group within the allotted time
1.11. Gamification & Engagement Module – Deliverable-Based WBS
1.11.1. Project Management & Documentation Deliverables
1.11.1.1. Software Project Management Plan (SPMP)
1.11.1.2. Software Requirements Specification (SRS)
1.11.1.3. Software Design Specification (SDS) & API Specification Document
1.11.1.4. Project Risk Log
1.11.1.5. Weekly Progress Reports
1.11.1.6. Git Repository
1.11.2. Backend & Database Deliverables
1.11.2.1. Points Economy Deliverables
1.11.2.1.1. Points Economy System

1.11.2.1.2. Points Database Schema & Ledger Structure (PostgreSQL)
1.11.2.1.3. Points Award APIs
1.11.2.1.4. Points Earning Logic
1.11.2.2. Badge & Achievement Deliverables
1.11.2.2.1. Badge & Achievement Rules Engine
1.11.2.2.2. Badge/Achievement Trigger Conditions
1.11.2.2.3. Automated Badge Awarding System
1.11.2.2.4. Achievement Tracking & History APIs
1.11.2.3. Leveling System Deliverables
1.11.2.3.1. Level Threshold Logic & XP Mapping
1.11.2.3.2. Level Calculation & Progression APIs
1.11.2.4. Challenges Module Deliverables
1.11.2.4.1. Challenge Schema (Daily/Weekly/Time-bound)
1.11.2.4.2. Challenge Creation & Progress Tracking System
1.11.2.4.3. Challenge Expiry & Completion Validation Logic
1.11.2.5. Admin Configuration Deliverables
1.11.2.5.1. Point Value Configuration API
1.11.2.5.2. Level Threshold Configuration API
1.11.2.5.3. Badge Rules Configuration API
1.11.2.5.4. Challenge Configuration API
1.11.3. Competitive & Reputation Deliverables
1.11.3.1. Leaderboard Deliverables
1.11.3.1.1. Leaderboard System
1.11.3.1.2. Leaderboard Ranking Algorithm with Tiebreaker Logic
1.11.3.1.3. Realtime Leaderboard Calculation API
1.11.3.1.4. Leaderboard Caching Strategy
1.11.3.2. Trust Score Deliverables
1.11.3.2.1. Trust Score Formula
1.11.3.2.2. Trust Score Calculation & Update Logic
1.11.3.2.3. Trust Score History
1.11.3.2.4. Trust Score API
1.11.3.3. Notification Deliverables
1.11.3.3.1. Notification Schema & Trigger Events
1.11.3.3.2. Notification Queue & Dispatcher
1.11.3.3.3. Notification Read/Archive APIs
1.11.4. Frontend Deliverables
1.11.4.1. User Dashboard Deliverables
1.11.4.1.1. Dashboard UI Components (Points, Level, XP Progress Bar)
1.11.4.1.2. Dashboard Integration with Points & Level APIs
1.11.4.1.3. Real-time Dashboard Data Refresh
1.11.4.2. Gamified Onboarding Deliverables
1.11.4.2.1. Interactive Multi-step Onboarding UI

1.11.4.2.2. Onboarding Step Completion Tracking System
1.11.4.2.3. Onboarding Reward Integration with Points Engine
1.11.4.3. Profile Data Deliverables
1.11.4.3.1. SVG Badge Icons/Assets
1.11.4.3.2. Profile Data Sync Logic
1.11.4.3.3. Badge Display Mapping (Top 3 Badges)
1.11.4.4. Challenges Panel Deliverables
1.11.4.4.1. Active/Completed Challenges UI
1.11.4.4.2. Challenges API Integration
1.11.4.4.3. Challenge Reward Claim Workflow
1.11.4.5. Notification Panel Deliverables
1.11.4.5.1. Notification Center UI
1.11.4.5.2. Notification API Integration
1.11.5. Integration & API Deliverables
1.11.5.1. API Deliverables
1.11.5.1.1. RESTful API Endpoints for Gamification Features
1.11.5.1.2. Authentication Middleware
1.11.5.1.3. Error Handling Mechanism
1.11.5.1.4. User Profile Data Endpoint
1.11.5.1.5. Optimized API Performance (<2 seconds response time)
1.11.5.1.6. Concurrent User Handling (500 users)
1.11.5.2. API Documentation Deliverables
1.11.5.2.1. Postman Collection
1.11.5.2.2. API Documentation (Request/Response Examples)
1.11.5.2.3. API Integration Guide
1.11.5.3. API Handover Deliverables
1.11.5.3.1. API Handover Session
1.11.5.3.2. API Documentation & Postman Collection
1.11.5.3.3. Integration Support & Clarifications
1.11.6. Finalization & Deployment Deliverables
1.11.6.1. End-to-End Tested System
1.11.6.2. Bug-Free & Polished UI
1.11.6.3. Code Repository & Documentation
1.12. Hardware And Asset Rental Integration
1.12.1. **No Submission from the Group within the allotted time**

WBS Dictionary:

WBS ID	Task Name	Description	Duration	Cost	Resources (# & Type)
1.4.1.1	Requirements Documentation	Identify functional & non-functional requirements for	2 days	0	2 (Team Lead + Backend Dev)

		matching, filtering, and recommendations			
1.4.1.2	Stakeholder Analysis Report	Identify stakeholders and their expectations	1 day	0	1 (Team Lead)
1.4.1.3	Project Documentation	Prepare planning documents (scope, WBS, schedule)	2 days	0	4 (All team members)
1.4.2.1	System Architecture	Design overall module architecture and data flow	3 days	0	2 (Backend + Frontend Dev)
1.4.2.2	Database Design (Skills, Projects, Matching Data)	Design schema for skills, projects, and matching data	2 days	0	1 (Backend Dev)
1.4.2.3	API Integration Design	Define APIs for Module 1, 2, and 3 communication	2 days	0	1 (Backend Dev)
1.4.3.1	Skill Matching Engine	Implement keyword-based skill comparison	2 days	0	2 (Backend Dev)
1.4.3.2	Budget & Rate Filtering System	Implement budget vs rate validation logic	1 day	0	1 (Backend Dev)
1.4.3.3	Match Percentage Calculation System	Calculate ranking score based on overlap	1 day	0	1 (Backend Dev)
1.4.3.4	Skill Gap Analysis System	Identify missing skills for partial matches	2 days	0	1 (Backend Dev)
1.4.3.5	Trending Skills Aggregation System	Aggregate frequently requested skills	2 days	0	1 (Backend Dev)

1.4.3.6	Notification System Integration	Send skill recommendations via Module 2	1 day	0	1 (Backend Dev)
1.4.4.1	Unit Test Results	Test individual components	1 day	0	1 (QA)
1.4.4.2	API Test Results	Validate API responses and data flow	1 day	0	2 (QA + Backend Dev)
1.4.4.3	Performance Test Results	Ensure response time under 2 seconds	1 day	0	1 (QA)
1.4.5.1	Integrated Module (API Level)	Integrate module via APIs with other modules	1 day	0	2 (Backend Dev + Team Lead)
1.4.5.2	Final Validated & Deployed System	Final testing and deployment readiness	2 days	0	2 (Team Lead + QA)
1.4.6.1	Requirements Management Plan (RMP)	Prepare Requirements Management Plan	1 day	0	1 (Team Lead)
1.4.6.2	Software Requirements Specification (SRS)	Document software requirements	2 days	0	4 (All team members)
1.4.6.3	Software Design Specification (SDS)	Document system design	2 days	0	2 (Backend + Frontend Dev)

WBS ID	Task Name	Description	Duration	Cost	Resources (# & Type)
1.6.1.1	Project Plan	Defines scope, schedule, and execution strategy	1 day	0	1 (Team Lead)
1.6.1.2	Meeting Records	Records meetings and coordination	Continuous	0	1 (Team Lead)

1.6.1.3	Progress Reports	Tracks module progress	Continuous	0	2 (Team Lead + QA)
1.6.2.1	SP-SRS	Functional & non-functional requirements	2 days	0	4 (All Team)
1.6.2.2	Requirements Analysis	Analyze and validate requirements	1 day	0	2 (Team Lead + Dev)
1.6.2.3	Requirements Documentation	Document finalized requirements	1 day	0	2 (Team Lead + Dev)
1.6.3.1	System Architecture	Overall system design and flow	3 days	0	2 (Backend + Frontend)
1.6.3.2	Database Schema	Design for messages, files, history	2 days	0	1 (Backend Dev)
1.6.3.3	API Design	Define APIs for communication	2 days	0	1 (Backend Dev)
1.6.3.4	UI/UX Design	Design interfaces and flows	2 days	0	1 (Frontend Dev)
1.6.4.1	Messaging System	Real-time one-to-one messaging	3 days	0	2 (Backend + Frontend)
1.6.4.2	Message Sync	Message delivery & synchronization	1 day	0	1 (Backend Dev)
1.6.5.1	Group Management	Create and manage groups	1 day	0	1 (Backend Dev)
1.6.5.2	Group Messaging	Multi-user chat system	2 days	0	2 (Backend + Frontend)
1.6.6.1	File Sharing	Upload and share files	2 days	0	2 (Backend + Frontend)
1.6.6.2	Secure Storage	Secure storage of files	1 day	0	1 (Backend Dev)
1.6.7.1	Notification System	Real-time alerts	1 day	0	1 (Backend Dev)
1.6.7.	Notification	Alert generation mechanism	1 day	0	1 (Backend

2	Trigger				Dev)
1.6.8.1	Link Generation	Generate meeting links	1 day	0	1 (Backend Dev)
1.6.8.2	Link Sharing	Share links in chat	1 day	0	1 (Backend Dev)
1.6.9.1	Message Storage	Store chat data	2 days	0	1 (Backend Dev)
1.6.9.2	Chat Retrieval	Retrieve chat history	1 day	0	1 (Backend Dev)
1.6.10.1	Chat UI	Messaging interface	2 days	0	1 (Frontend Dev)
1.6.10.2	Notification UI	Alerts interface	1 day	0	1 (Frontend Dev)
1.6.10.3	Interaction Flows	User navigation design	1 day	0	1 (Frontend Dev)
1.6.11.1	Messaging APIs	Backend communication services	2 days	0	1 (Backend Dev)
1.6.11.2	User Integration	Integration with user module	1 day	0	1 (Backend Dev)
1.6.11.3	Module Integration	Integration with other modules	2 days	0	2 (Backend + QA)
1.6.12.1	Unit Testing	Test individual components	1 day	0	1 (QA)
1.6.12.2	Integration Testing	Test module interactions	2 days	0	2 (QA + Backend)
1.6.12.3	System Testing	End-to-end testing	2 days	0	1 (QA)
1.6.12.4	Bug Logs	Record and fix issues	Continuous	0	2 (QA + Dev)
1.6.13.1	SP-SRS Doc	Final requirements document	2 days	0	4 (All Team)

1.6.13.2	SP-SDS Doc	Final design document	2 days	0	2 (Dev Team)
1.6.13.3	Technical Docs	Implementation documentation	2 days	0	2 (Dev Team)
1.6.14.1	Deployment	Deploy module	1 day	0	2 (Backend + QA)
1.6.14.2	Demonstration	Final presentation	1 day	0	4 (All Team)
1.6.14.3	Final Report	Complete documentation	2 days	0	4 (All Team)

WBS Code	Work Package	Description	Duration	Dependencies	SRS Requirement Trace
1.11.1.1	Software Project Management Plan (SPMP)	Complete project management plan with all sections	4	None	-
1.11.1.2	Software Requirements Specification (SRS)	Complete software requirements specification document	4	1.11.1.1	-
1.11.1.3	Software Design Specification (SDS) & API Specification	Design document and API contracts with Modules 1, 3, 9, 13	6	1.11.1.2	REQ-46
1.11.1.4	Project Risk Log	Create risk register with mitigation strategies	2	None	-
1.11.1.5	Weekly Progress Reports	Status updates to instructor and integration group	6	None	-

1.11.1.6	Git Repository	Repository setup with branching strategy	2	None	OR-5
1.11.2.1	Points Database Schema & Ledger	Create UserProgress and PointsLedger models	3	1.11.1.6	REQ-2, REQ-3
1.11.2.2	Points Award APIs	Implement /api/points/award endpoint	6	1.11.2.1.1	REQ-1
1.11.2.3	Points Earning Logic	Atomic updates with race condition prevention	5	1.11.2.1.2	NFR-6, NFR-7
1.11.2.4	Badge Trigger Conditions	Define badge conditions	2	1.11.2.1.1	REQ-14
1.11.2.5	Automated Badge Awarding System	Rule evaluation and badge assignment	6	1.11.2.2.1	REQ-12, REQ-13, REQ-15
1.11.2.6	Achievement Tracking APIs	Implement badge retrieval APIs	4	1.11.2.2.2	REQ-13
1.11.2.7	Level Threshold Logic	Define levels based on points	1	None	REQ-7, REQ-10
1.11.2.8	Level Progression APIs	Implement level calculation and progression APIs	5	1.11.2.3.1	REQ-8, REQ-11
1.11.2.9	Challenge Schema	Design daily/weekly challenge structure	2	1.11.2.1.1	REQ-17, REQ-18
1.11.2.10	Challenge Progress Tracking	Track user challenge progress	4	1.11.2.4.1	REQ-19

1.11.2.11	Challenge Reset Logic	Implement recurring reset schedulers	4	1.11.2.4.2	REQ-21
1.11.2.12	Point Value Configuration API	Configure point values	2	1.11.2.1.1	REQ-5
1.11.2.13	Level Threshold Configuration API	Configure level thresholds	2	1.11.2.3.1	REQ-10
1.11.2.14	Badge Rules Configuration API	Configure badge rules	2	1.11.2.2.1	REQ-16
1.11.2.15	Challenge Configuration API	Configure challenges	2	1.11.2.4.1	REQ-22
1.11.3.1	Leaderboard Algorithm	Ranking with tiebreaker logic	3	1.11.2.1.1	REQ-23, REQ-24
1.11.3.2	Leaderboard API	Fetch leaderboard rankings	5	1.11.3.1.1	REQ-25, REQ-26, REQ-27
1.11.3.3	Leaderboard Caching	Optimize performance using cache	4	1.11.3.1.2	NFR-3
1.11.3.4	Trust Score Formula	Define calculation formula	2	None	REQ-33, REQ-36
1.11.3.5	Trust Score Calculation	Implement calculation logic	4	1.11.3.2.1	REQ-34
1.11.3.6	Trust Score History	Track score changes	2	1.11.3.2.2	-
1.11.3.7	Trust Score API	Provide trust score endpoint	2	1.11.3.2.2	REQ-35

1.11.3.8	Notification Schema	Design notification data structure	2	1.11.2.1.1	REQ-38
1.11.3.9	Notification Generation	Implement queue/dispatcher	5	1.11.3.3.1	REQ-37
1.11.3.10	Notification APIs	Implement notification endpoints	3	1.11.3.3.2	REQ-39, REQ-40, REQ-41
1.11.4.1	Dashboard UI Components	Points, Level, XP progress bar	4	None	REQ-6, REQ-11
1.11.4.2	Dashboard API Integration	Connect dashboard with backend APIs	4	1.11.4.1.1	-
1.11.4.3	Real-time Data Refresh	Live updates via polling/WebSocket	2	1.11.4.1.2	-
1.11.4.4	Onboarding UI	Multi-step onboarding interface	4	None	REQ-28
1.11.4.5	Onboarding Tracking	Track completion of steps	4	1.11.4.2.1	REQ-29, REQ-32
1.11.4.6	Onboarding Reward Integration	Points for onboarding steps	4	1.11.4.2.2	REQ-30, REQ-31
1.11.4.7	Badge Assets	SVG badge icons	3	None	-
1.11.4.8	Profile Data Sync	Sync with Module 1	3	1.11.2.2.3	REQ-35
1.11.4.9	Badge Mapping Logic	Top 3 badges display	2	1.11.4.3.2	-
1.11.4.10	Challenges UI	Display active/completed challenges	3	None	REQ-17, REQ-18

1.11.4.11	Challenges API Integration	Fetch and show progress	3	1.11.4.4.1	REQ-19
1.11.4.12	Challenge Reward Workflow	Claim rewards	2	1.11.4.4.2	REQ-20
1.11.4.13	Notification UI	Notification center	2	None	REQ-39
1.11.4.14	Notification API Integration	Fetch & mark notifications	4	1.11.4.5.1	REQ-40, REQ-41